

Gear System Hand Calculations

Assumptions

- At least 6 teeth must be engaged with the belt at all times ($N_{\text{engaged}} > 6$)
- The gear ratio should be lower than 5:1
- Timing belt transmits torque instantaneously; thus pinion torque is a function of driven gear torque and gear ratio
- Driven gear is modeled as cylinder for inertia calculations
- LiDAR unit is modeled as shaft for inertia calculations

Diagram

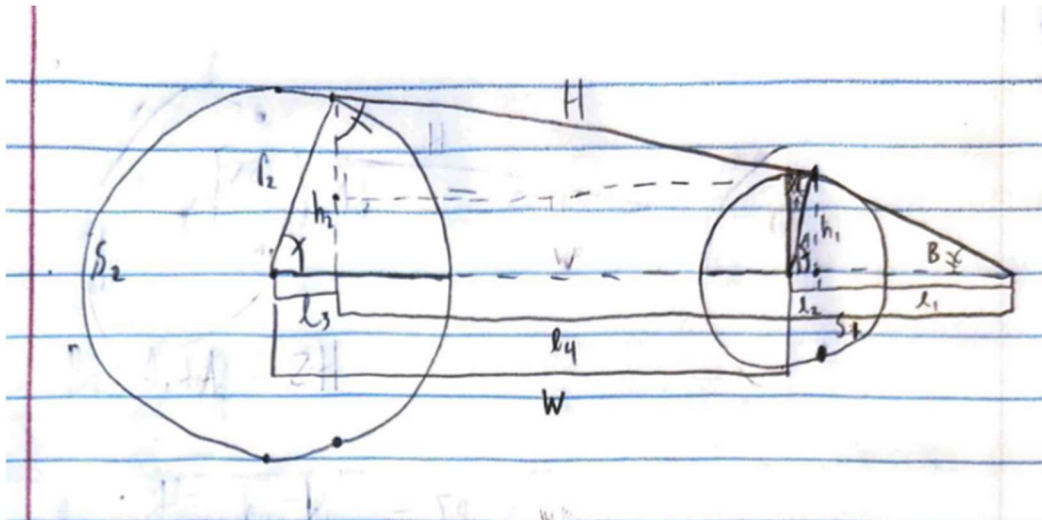


Figure J1. Hand Sketch of Gear Geometry

Calculations

$$N_{\text{Pinion}} = 15 \text{ teeth}$$

$$P = 2 \text{ mm}$$

$$\text{Ratio} = 5:1$$

$$D_{\text{Pinion}} = 9.7 \text{ mm}$$

$$D_{\text{Gear}} = 48.5 \text{ mm}$$

$$r_1 = \frac{D_{\text{Pinion}}}{2} = 4.85 \text{ mm}$$

$$r_2 = \frac{D_{\text{Gear}}}{2} = 24.25 \text{ mm}$$

$$N_{\text{Engaged}} \geq 6 \text{ teeth}$$

$$\angle A = \frac{6}{15} * \frac{360^\circ}{2} = 72^\circ$$

$$\angle B = 90^\circ - 72^\circ = 18^\circ$$

$$h_1 = r_1 \sin(72^\circ)$$

$$h_2 = r_2 \sin(72^\circ)$$

$$l_1 = h_1 \tan(72^\circ)$$

$$l_2 = r_1 \cos(72^\circ)$$

$$l_3 = r_2 \cos(72^\circ)$$

$$l_4 = h_2 \tan(72^\circ)$$

$$W > l_4 + l_3 - l_1 - l_2 \\ = \mathbf{62.78mm}$$

$$S_1 = \frac{6}{15} \pi D_{\text{Pinion}}, S_2 = \frac{9}{15} \pi D_{\text{Gear}}$$

$$H = \frac{h_2 - h_1}{\cos(72^\circ)}$$

$$L_{\text{Belt}} > S_1 + S_2 + 2H \\ = \mathbf{223.0285 \text{ mm}}$$

LiDAR

$$m = 0.692 \text{ kg}$$

$$R = 0.0435 \text{ m}$$

$$I_L = \frac{1}{12} MR^2 =$$

$$7.916 * 10^{-5} \text{ kgm}^2$$

Driven Gear

$$m = 0.0189 \text{ kg}$$

$$R = 0.024 \text{ m}$$

$$I_G = 0.545 * 10^{-5} \text{ kgm}^2$$

$$\omega_0 = 0 \text{ rad/s}, t = 2s$$

$$\omega_1 = 14 \frac{\text{rad}}{\text{s}} = 7 \text{ Hz}$$

$$\alpha = \frac{\omega_1 - \omega_0}{t} = 7\pi \text{ rad/s}^2$$

$$\tau_G = (I_L + I_G) * \alpha = 3.721 \text{ mNm}$$

Torque and RPM of Drive Gear

$$\tau_P = \frac{3.721 \text{ mNm}}{5} = \mathbf{0.7442 \text{ mNm}}$$

$$\omega_P = 7 \text{ Hz} * 5 = 35 \text{ Hz} \\ = \mathbf{2100 \text{ Rpm}}$$